

EE6406 – Analytic and Ensemble Machine Learning

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- L1: Introduction (LZP)
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- L6: Advanced Analytic Classification (TKA)
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- L10: Classical Ensemble Algorithms – Random Forest, Adaboost, Gradient Boosting (S. Liu)
- L11: Advanced Boosting Algorithms – XGBoost and LightGBM (S. Liu)
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- L13: Industrial Applications – End-to-end Ensemble Models (S. Liu)

Lecture 1 - Introduction

Learning Objectives

- Explain the motivation for studying machine learning.
- Describe the roles of analytic learning and ensemble learning in modern AI.
- Understand how this course is organized.
- Identify the main topics covered in the remaining lectures.

EE6406 – Final exam and CA's

- Final exam 60%:
4 questions: 2.5 questions from Part 1 (Weeks 1 – 8);
1.5 questions from Part 2 (Weeks 9 - 13)
- CA (quizzes only) 40%:
 - CA1 – Part I: online quiz, 10%, **Week 5** (in class, closed-book)
 - CA2 – Part I: online quiz, 15%, **Week 8** (in class, closed-book)
 - CA3 – Part II: online quiz, 15%, **Week 12** (in class, closed-book)

Text Book:

[1] “Analytic Learning Methods for Pattern Recognition” by K.-A. Toh, H. Zhuang, Simon Liu and Z. Lin, Springer, 2025, Springer (NTU Lib., online)

References:

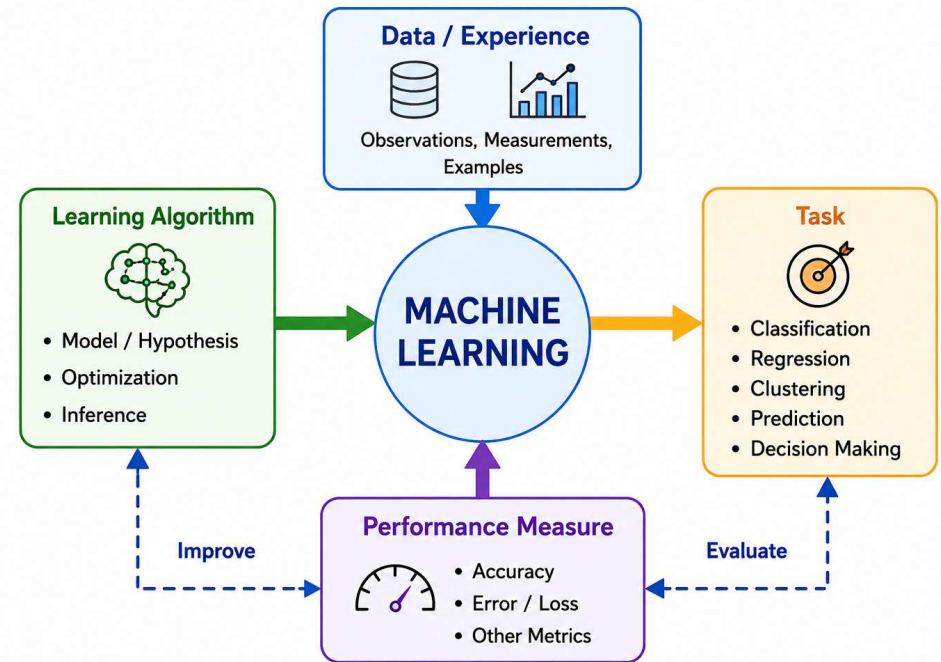
- [2] “Combining Pattern Classifiers: Methods and Algorithms” (2nd edition) by Ludmila I. Kuncheva, Wiley, ISBN: 978-1-118-31523-1, 2014.
- [3] “The Elements of Statistical Learning: Data Mining, Inference and Prediction” (2nd edition) by Trevor Hastie, Robert Tibshirani and Jerome Friedman, Springer, New York, 2017.
- [4] “Machine Learning”, Tom M. Mitchell, (1997).
- [5] Tianqi Chen and Carlos Guestrin, “XGBoost: A scalable tree boosting system,” In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785-794, ACM, 2016.
- [6] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu, “LightGBM: A highly efficient gradient boosting decision tree,” In Advances in Neural Information Processing Systems 30, pp. 3149-3157, NIPS, 2017.

What Is Machine Learning?

Machine learning is a field of artificial intelligence that enables computers to learn from data and experience rather than being explicitly programmed for every task.

A widely accepted definition was given by Tom M. Mitchell (1997):

“A computer program is said to learn from **experience E** with respect to some **task T** and **performance measure P** if its performance at tasks in T, as measured by P, **improves** with experience E.”



Key elements

Experience (E): Training data or interactions

Task (T): Classification, regression, clustering, etc.

Performance (P): Accuracy, error rate, F1-score, reward, etc.

A Simple Example: Spam Email Classification

Suppose we want a computer to automatically classify an incoming email as

Spam vs. Non-spam (Ham)

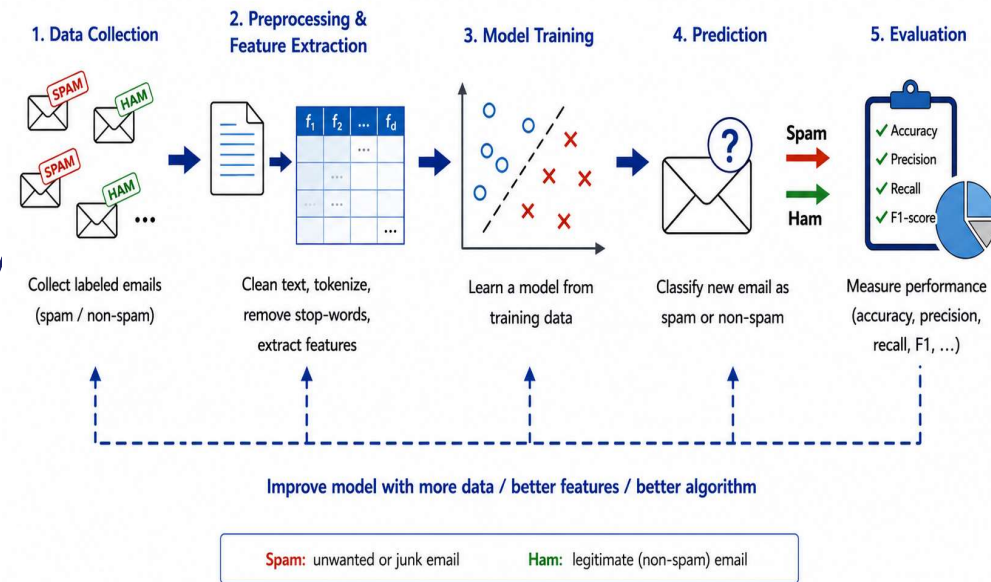
Instead of explicitly programming the classification rules, we allow the computer to **learn from previously labelled emails**.

Machine Learning Workflow

- Collect labelled emails
- Extract useful features
- Learn a classification model
- Classify new emails
- Evaluate the performance

Learning from data is the central idea behind machine learning.

Example: Spam Email Classification Pipeline



Key observations

- The computer learns from **examples**.
- Higher data quality often leads to better classification performance.
- The same learning framework applies to many other applications.

The approaches to machine learning can generally be classified into supervised learning, unsupervised Learning, semi-supervised learning, and reinforcement learning.

1. Supervised Learning		2. Unsupervised Learning		3. Semi-supervised Learning		4. Reinforcement Learning	
							
Definition	Learn a mapping from inputs to known output labels using labeled data.	Definition	Discover structure or patterns in unlabeled data without using known output labels.	Definition	Learn a model using a small amount of labeled data together with a large amount of unlabeled data.	Definition	Learn to make decisions by interacting with an environment and receiving rewards or penalties.
Goal	Minimize prediction error on unseen data.	Goal	Reveal hidden structure, groupings, or low-dimensional representations.	Goal	Improve learning performance when labeled data is scarce.	Goal	Maximize cumulative (long-term) reward.
Training Data	Labeled data: $\{(x_i, y_i)\}$	Training Data	Unlabeled data: $\{x_i\}$	Training Data	Combination of labeled and unlabeled data.	Interaction	Agent interacts with environment: state $\rightarrow$ action $\rightarrow$ reward $\rightarrow$ next state.
Output/Result	Predict labels (classification) or values (regression) for new inputs.	Output/Result	Clusters, associations, or representations.	Output/Result	Predict labels or values for new inputs.	Output/Result	Learn a policy that maps states to actions.
Typical Tasks	Classification, Regression	Typical Tasks	Clustering, Dimensionality Reduction, Association Rule Mining	Typical Tasks	Classification, Regression	Typical Tasks	Sequential decision making, control, robotics, game playing

The Evolution of Machine Learning

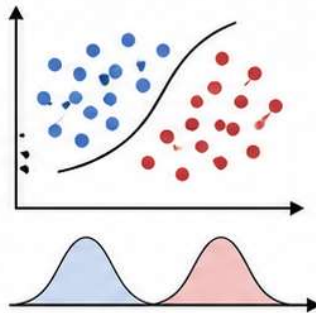
Increasing ability to learn from data

1 Knowledge-driven (Rule-based AI)



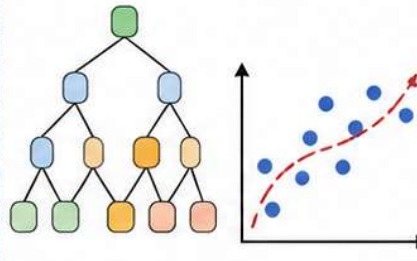
Main idea:
Human experts encode knowledge as rules. Systems follow explicit instructions.

2 Statistics-driven (Pattern Recognition)



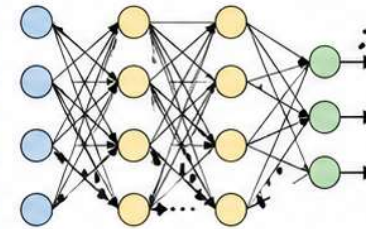
Main idea:
Use statistics and probability to model data and make inferences. Focus on patterns.

3 Data-driven (Machine Learning)



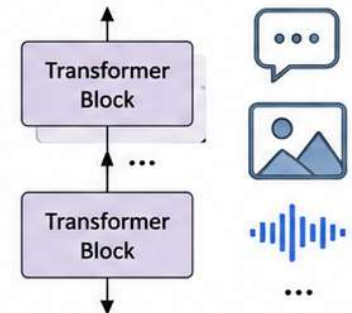
Main idea:
Learn models from data that can generalize to unseen examples. Emphasis on algorithms and optimization.

4 Representation-driven (Deep Learning)



Main idea:
Learn hierarchical representations directly from large amounts of data using deep neural networks.

5 Foundation Models (LLMs / Multimodal AI)

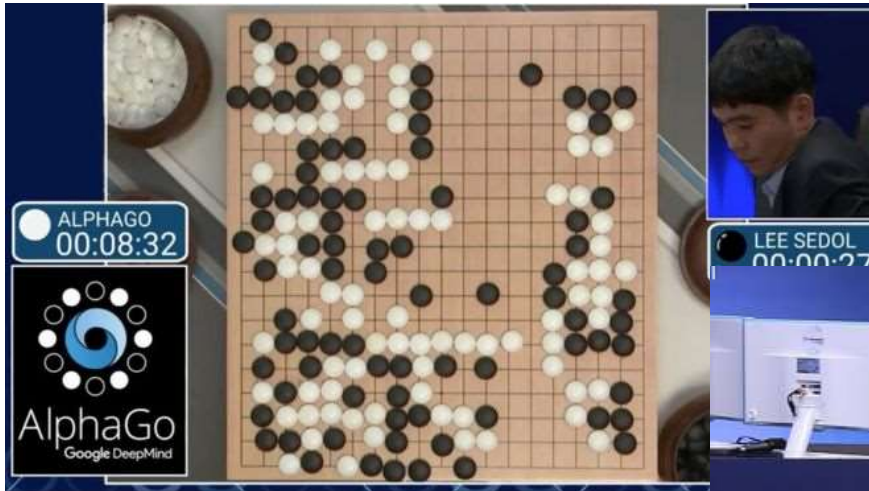


Main idea:
Pretrain large models on massive data (text, image, audio, etc.) and adapt them to many tasks. Strong general capability.



Key message: Machine learning has evolved from relying on human-designed knowledge toward **automatically learning** increasingly **powerful representations** from data.

Breakthrough in Machine Learning (AI) - AlphaGo



<https://www.bbc.com/news/technology-35785875>

March 2016, AlphaGo – an artificial intelligence program developed by Google DeepMind – played a best-of-five Go match against legendary South Korean grandmaster Lee Sedol in Seoul, South Korea.



<https://www.businessinsider.com/googles-alphago-made-artificial-intelligence-history-2016-3>

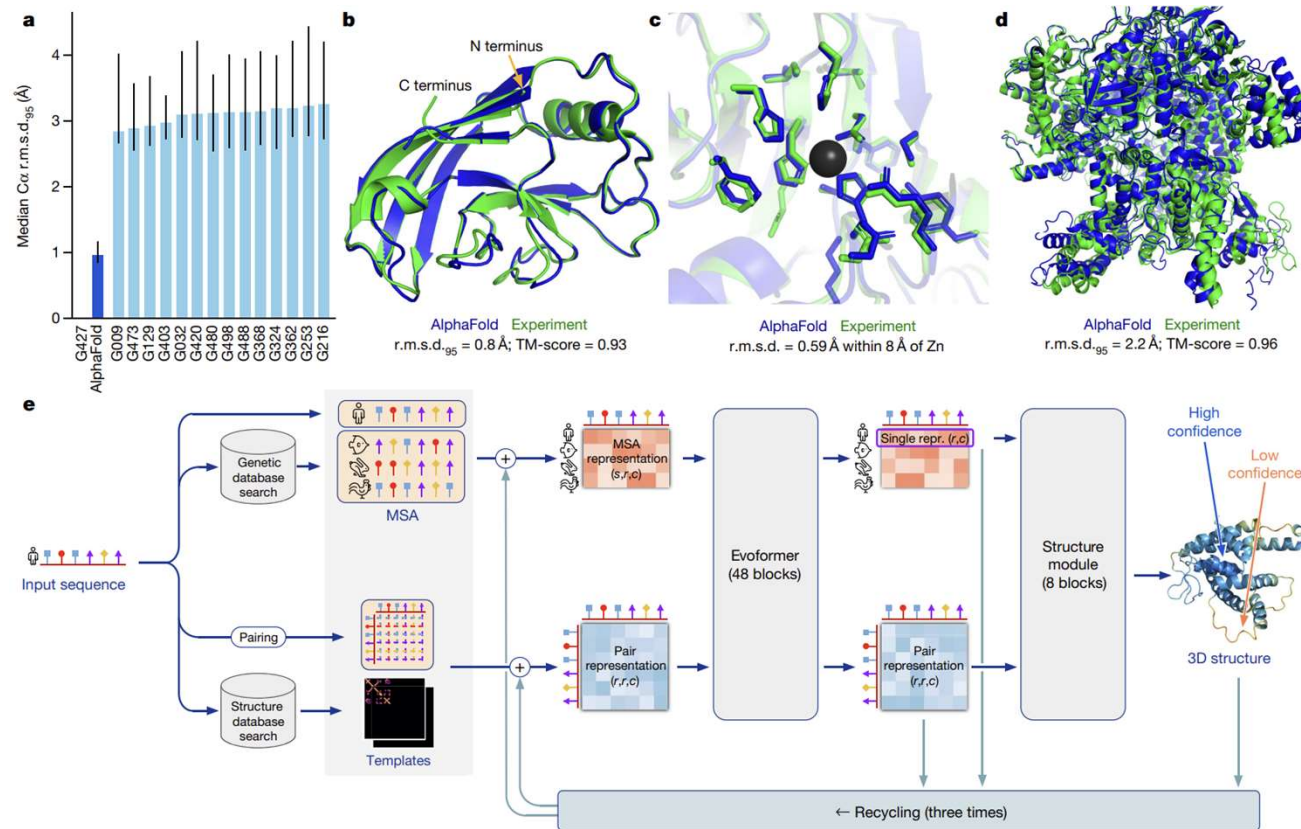
Why it was a breakthrough?

- Go is an ancient, extremely complex strategy board game with a vast number of possible positions – far more than chess – making it long believed that computers wouldn't master it soon. (CNBC 2017/05/23)
- Unlike earlier AI like IBM's Deep Blue (which beat the world chess champion in 1997), AlphaGo didn't rely on brute-force calculation but used deep learning and reinforcement learning, enabling it to learn from playing millions of games and to “intuitively” evaluate strong moves. (Forbes 2018/02/07)

Global Impact

- The results shocked the AI and Go communities, because the consensus had been that it would take many more years before AI could defeat a top human Go player. (People's Daily 2016/05/28)
- The match was widely covered by international media and sparked interest in AI research, machine learning, and how these technologies can extend beyond games to real-world problems.
- Recent development. July 2026 - World No. 1 human professional Shin Jin-seo played with Go AI KataGo in a historic three-game handicap match.

Breakthrough in Machine Learning (AI) - AlphaFold



<https://www.nature.com/articles/s41586-021-03819-2.pdf>

What the breakthrough was

- AlphaFold is an artificial intelligence system developed by DeepMind that can predict a protein's 3D structure directly from its amino acid sequence with very high accuracy. On average, DeepMind's AI system successfully predicted proteins' three-dimensional shapes to within the width of about one atom. The CASP organizers themselves declared that the protein folding problem had been solved. (Forbes 2021/12/10)
- In 2020 at the CASP14 competition (Critical Assessment of Protein Structure Prediction), AlphaFold dramatically outperformed all other methods and reached levels of accuracy comparable to laboratory experiments— so precise that the organizers said it effectively solved the protein-folding problem. (Nature 2023/03/14)

Number of Protein Structures

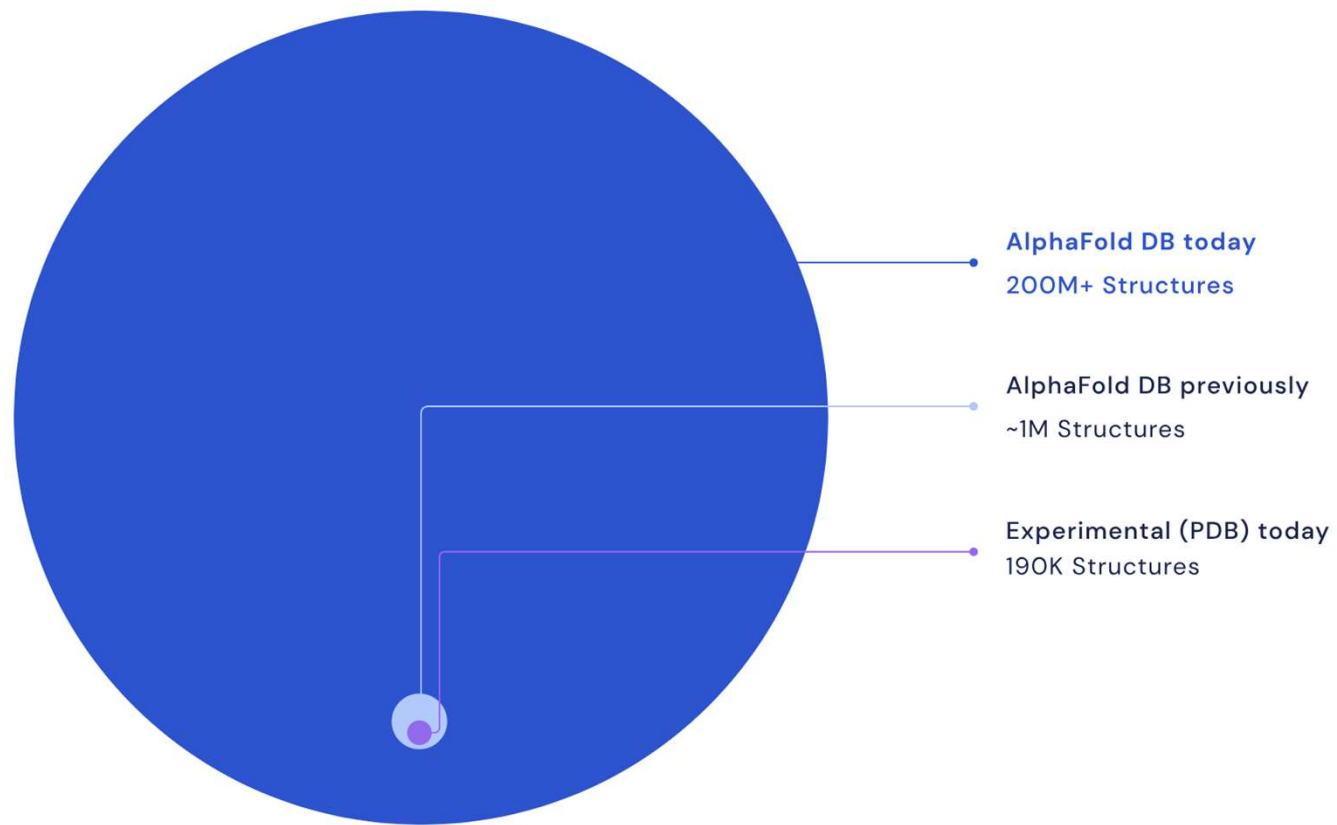


Image source (2022): <https://www.deepmind.com/blog/alphafold-reveals-the-structure-of-the-protein-universe>

Why it matters?

- Protein structure is central to biology: A protein's function depends on its 3D shape -- which historically could only be found through slow, costly experimental methods like X-ray crystallography or cryo-EM. (Google DeepMind)
- Huge leap in accuracy and speed: AlphaFold's predictions are often within the width of a single atom of the real structure -- a level orders of magnitude better than previous computational methods. (Forbes 2021/12/10)

Impact on science and medicine

- AlphaFold's success has transformed biomedical research: Protein structure prediction is now high-throughput -- what once took years experimentally can be done in hours or less. In 2021--22, the AlphaFold Protein Structure Database was released, with hundreds of millions of protein structures freely available worldwide. (Google DeepMind)
- Researchers are using predictions to study disease mechanisms, design better drugs, improve enzyme engineering, and speed biological discovery. (NIH 2024/11/09)

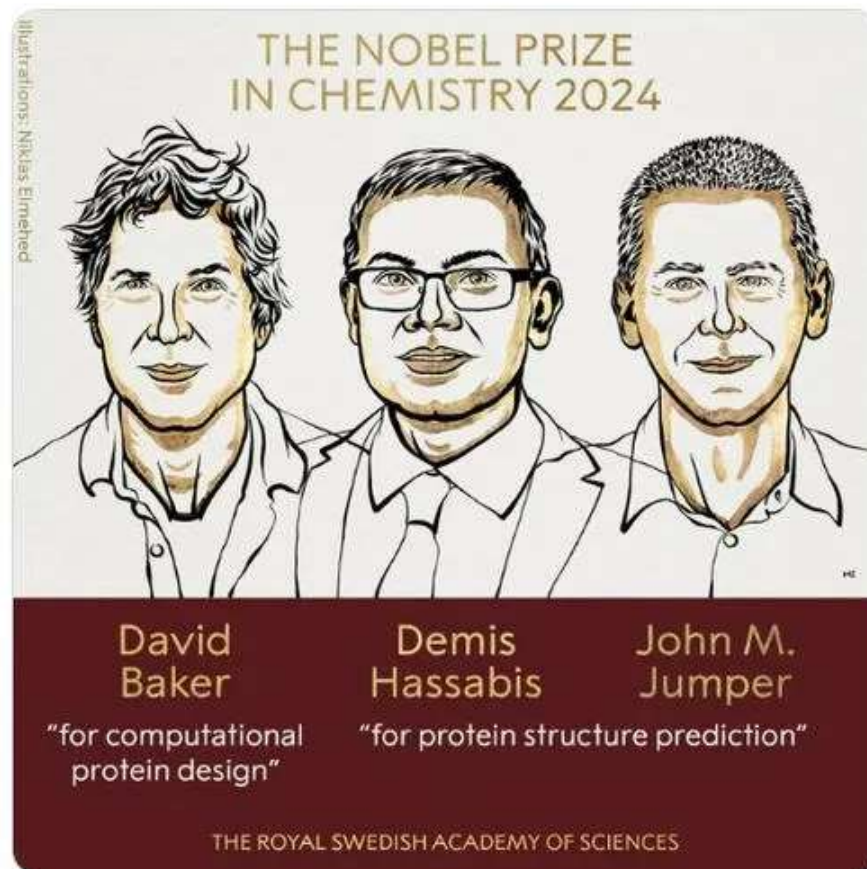


The Nobel Prize @NobelPrize · 17秒

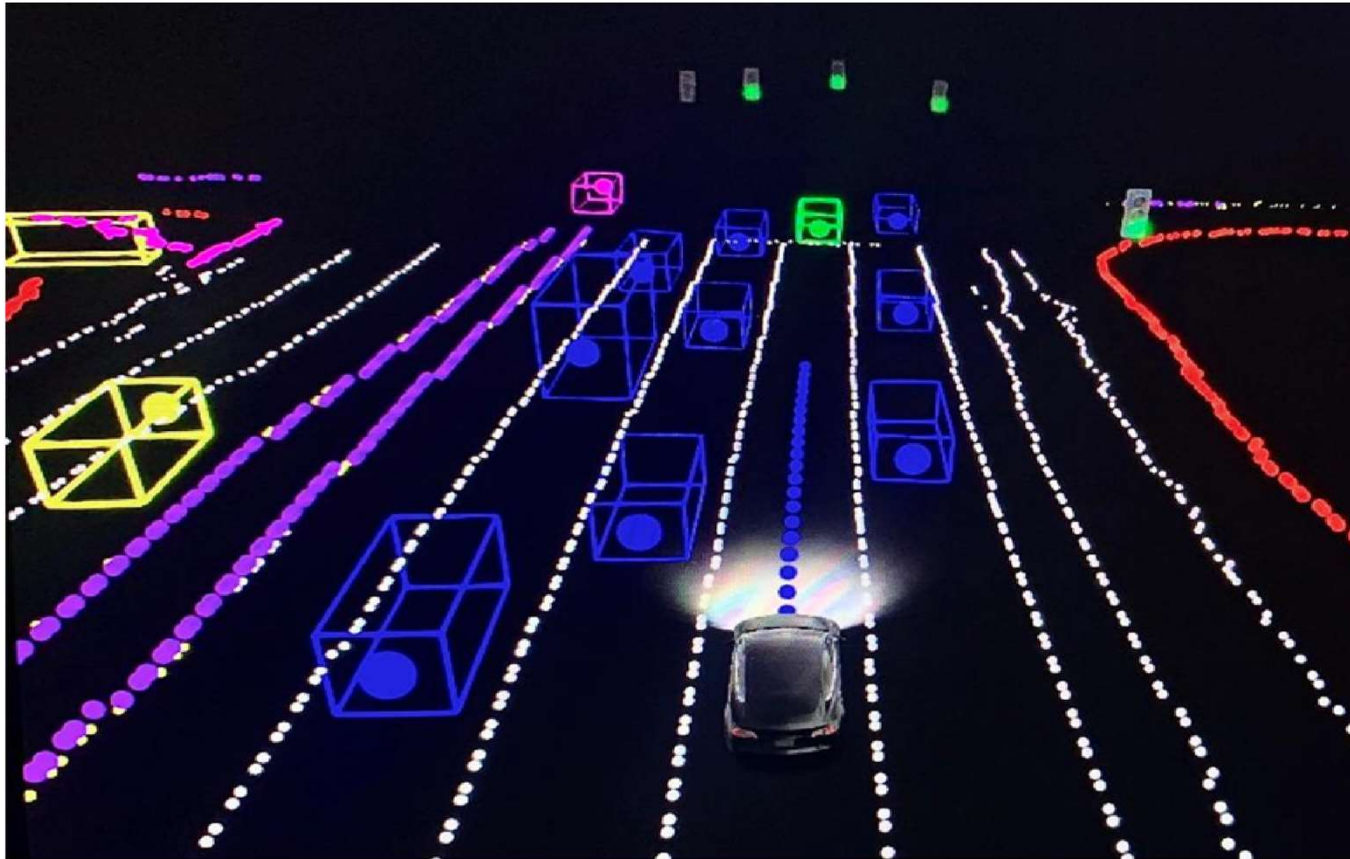
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BREAKING NEWS

The Royal Swedish Academy of Sciences has decided to award the 2024 [#NobelPrize](#) in Chemistry with one half to David Baker “for computational protein design” and the other half jointly to Demis Hassabis and John M. Jumper “for protein structure prediction.”



An image of how Tesla's full self driving (FSD) artificial intelligence sees the world. (TESLA)



<https://www.trtworld.com/magazine/five-recent-breakthroughs-in-ai-you-probably-missed-57056>

- Full Self-Driving (FSD) is Tesla's flagship advanced driver-assistance system designed to enable vehicles to navigate, steer, change lanes, handle intersections, and drive through city and highway environments with minimal human input -- moving toward true autonomous driving.
- It builds on earlier Autopilot systems and is offered as a paid upgrade or subscription.

(SCMP 2024/05/09)

Breakthrough Features & Capabilities

- Advanced navigation: FSD can handle complex highway merges, lane changes, city turns, and stop-controlled intersections with supervised autonomy.
- End-to-end AI: Tesla's neural networks interpret raw image data to directly output steering, braking, and acceleration actions – a paradigm shift from rule-based decision systems.
- Constant fleet learning: Real-world driving data from millions of vehicles continuously feeds model improvements, enabling rapid iteration and refinement.

(SCMP 2024/05/09)

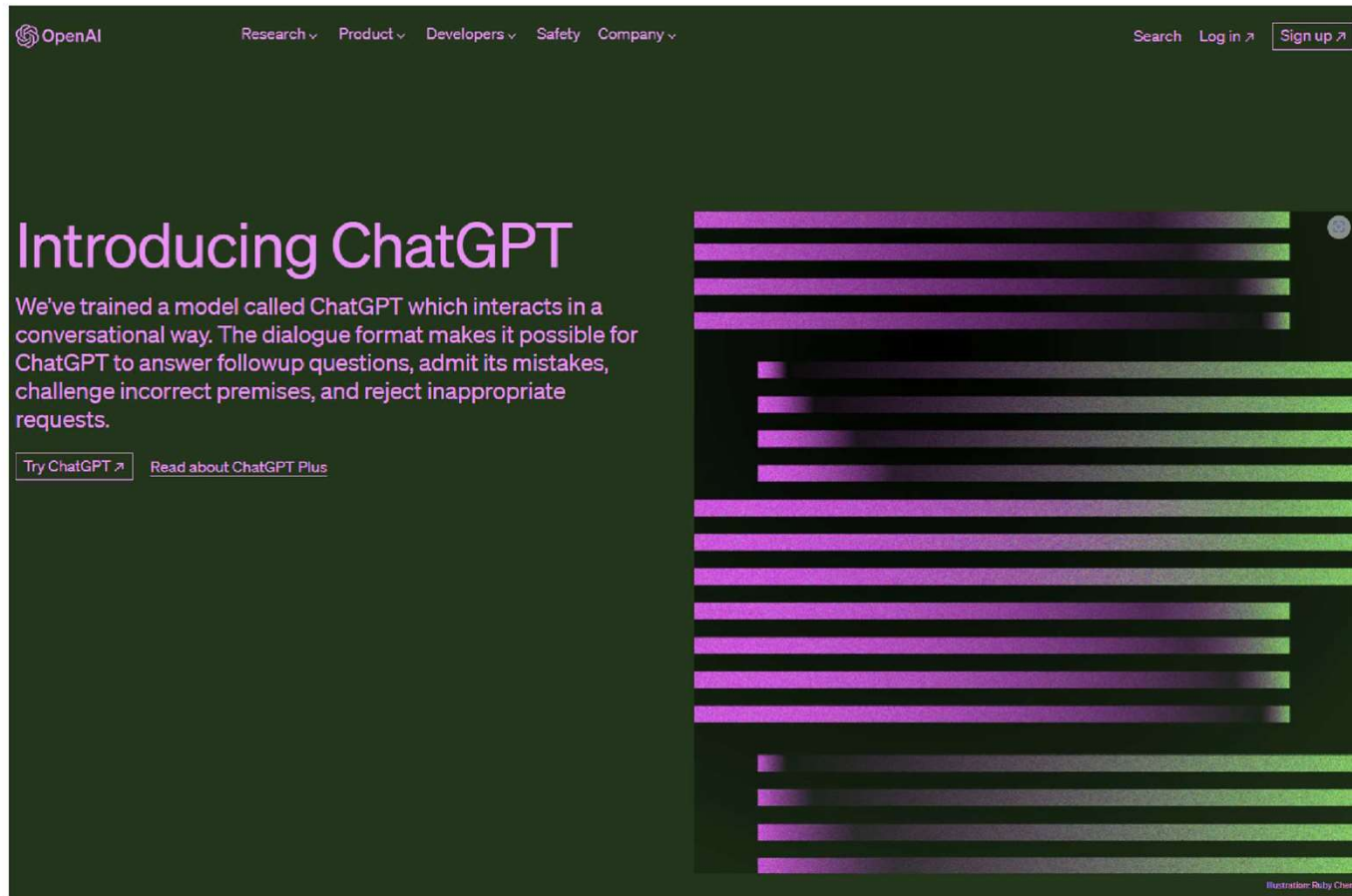


Image source: <https://openai.com/index/chatgpt/>

ChatGPT was first introduced to the public on November 30, 2022.

Why ChatGPT Was a Turning Point?

Before ChatGPT:

- AI systems were specialized
- Interaction required technical expertise
- Capabilities were fragmented

After ChatGPT:

- AI became conversational, general, and adaptive
- Language became the primary human–AI interface
- AI shifted from a backend tool to a frontline productivity partner

What Made these Happen?

- Massive amount of data
- Hardware: CPU, GPU and TPU
- Software: Open Source Platform and Ecosystem
- Algorithms: Advanced Algorithms and Model Architectures
- Feedback Loops and Continuous Learning

Massive Amounts of Data

Why it matters: Data is the raw material of learning.

- Large, diverse datasets allow models to generalize rather than memorize.
- Real-world data (images, text, protein sequences, driving logs) captures complexity that handcrafted rules cannot.
- Self-play and simulation (e.g., AlphaGo) and fleet data (e.g., Tesla) multiply effective data volume.
- Data scale often correlates directly with performance improvements.

Without large-scale data, modern deep learning would not work.

Hardware: CPU, GPU and TPU

Why it matters: Training large models is computationally intensive.

- GPUs enabled massive parallel computation, making deep neural networks practical.
- TPUs and AI accelerators further optimized matrix operations used in deep learning.
- Hardware improvements reduced training time from months to days or hours.
- Energy efficiency gains made large-scale AI economically viable.

Hardware turned theoretical models into deployable systems.

Software: Open Source Platform and Ecosystem

Why it matters: Innovation scales when tools are shared.

- Frameworks like TensorFlow, PyTorch, CUDA, and scientific libraries lowered barriers to entry.
- Open datasets, benchmarks, and pretrained models accelerated progress.
- Researchers and engineers worldwide could replicate, improve, and combine ideas rapidly.
- Industry-academia feedback loops sped up innovation.

Open ecosystems transformed AI from isolated research into a global movement.

Algorithms: Advanced Algorithms & Model Architectures

Why it matters: Scale alone is not useful without the right learning methods.

- Breakthroughs in deep neural networks, transformers, reinforcement learning, and self-supervised learning unlocked new capabilities.
- End-to-end learning replaced brittle rule-based systems.
- Architectural innovations improved stability, efficiency, and accuracy.
- Algorithms learned representations directly from data instead of relying on human-engineered features.

Algorithmic innovation turned data and compute into intelligence.

Feedback Loops & Continuous Learning

Why it matters: Breakthrough systems improve after deployment.

- Models learn from their own outputs (self-play, simulation).
- Real-world usage feeds continuous retraining.
- Performance gains compound over time rather than plateau.

AI systems now evolve dynamically, not statically.

Why This Machine Learning Course?

Machine learning has become one of the most influential technologies of the 21st century.

Today it powers:

- intelligent search
- speech recognition
- autonomous driving
- medical diagnosis
- scientific discovery
- ...

It also faces important challenges:

- increasing computational cost
- large training datasets
- limited interpretability
- high energy consumption
- extensive parameter tuning
- ...

This course introduces two complementary paradigms —
Analytic Learning and **Ensemble Learning**
that address some of these challenges.

Why Analytic Learning and Ensemble Learning?

Despite tremendous success, many machine learning methods still rely on

- iterative optimisation
- large labelled datasets
- extensive parameter tuning
- high computational cost
- limited interpretability

This motivates alternative learning paradigms

This course focuses on

Analytic Learning and Ensemble Learning

which can improve

- learning efficiency
- interpretability
- scalability (mainly for ensemble learning currently)
- robustness

The remainder of this course explores these two complementary paradigms.

What is Analytic Learning?

https://link.springer.com/referenceworkentry/10.1007/978-1-4419-1428-6_490

- The word analytic comes from the Greek word 'analytikos', which means having the ability to analyze and divide the whole into its components or elements.
- Analytic Learning is an analytical approach to learning that uses prior knowledge as a base from which concepts can be described, hypotheses can be developed, and concepts can be rationally generalized by analyzing the components and the structure of the concepts.
- Analytic learning allows the learner to process information, break it into its component parts, and generate hypotheses by using critical and logical thinking skills.

The remainder of this course explores these two complementary paradigms.

Why Analytic Learning?

Most modern machine learning algorithms are trained by iterative optimization. Examples include Gradient Descent (GD), Stochastic Gradient Descent (SGD), Adam, and related optimization methods.

Training may require hundreds or thousands of iterations before convergence.

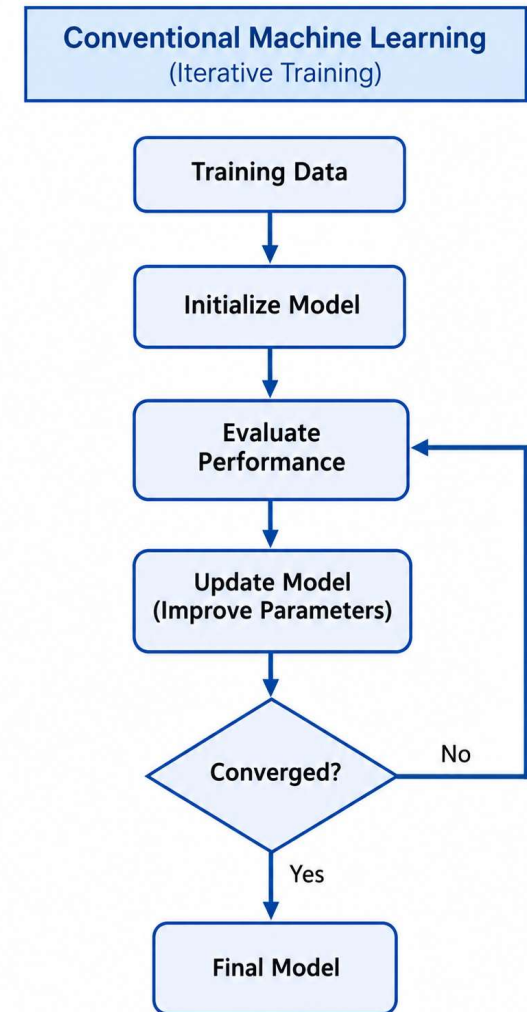
Performance depends on many hyperparameters: learning rate

- initialization
- batch size
- optimizer selection

Convergence is not always guaranteed.

Large-scale problems may require substantial computational resources.

Although iterative optimization is powerful, it is not always the most efficient learning strategy.

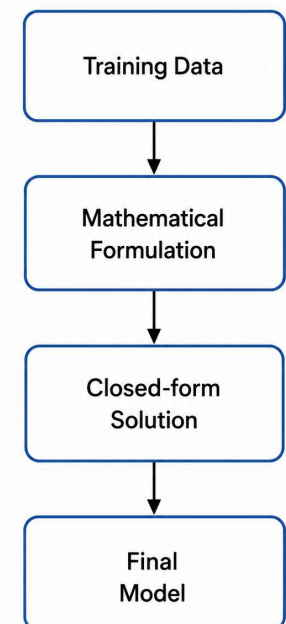


Can we obtain the model directly without performing iterative optimization?

- Analytic learning computes model parameters using closed-form mathematical solutions. No iterative optimization is required in many learning problems. The solution is obtained through linear algebra and optimization theory. Benefits include:
 - deterministic solution
 - significantly faster training for small learning components
 - no learning-rate tuning
 - improved reproducibility
 - particularly attractive for small and medium-scale real-time applications.

Instead of "learning by repeated trial-and-error," analytic learning aims to "solve the learning problem directly."

Analytic Learning



Feature	Conventional (Iterative) Learning	Analytic Learning
Training process	Iterative updates	Direct (closed-form) solution
Learning rate	Required	Not required
Initialization	Important	Usually unnecessary
Convergence	May be slow or need tuning	Immediate (if solution exists)
Deterministic solution	Not always	Usually yes
Hyperparameter tuning	Often many	Typically none or few
Computational cost	Often high	often much lower for small learning components
Scalability	suitable for medium- and large-scale problems	often better for small problems
Sensitivity to data noise	Can be high	Often more robust
Interpretability	Often lower	Often higher
Typical methods	GD, SGD, Adam, Backpropagation, etc.	Least Squares, LDA, Ridge Regression, Analytic RVM, Closed-form Classifiers, etc.

Closed-form Learning

A closed-form solution computes the model parameters directly using mathematical formulas, rather than through repeated iterative updates.

Typical mathematical tools include:

- Linear algebra
- Matrix decomposition
- Least-squares estimation
- Regularization
- Convex optimization

Examples:

- Linear regression
- Ridge regression
- Kernel ridge regression
- Recursive classification TER learning
- ...

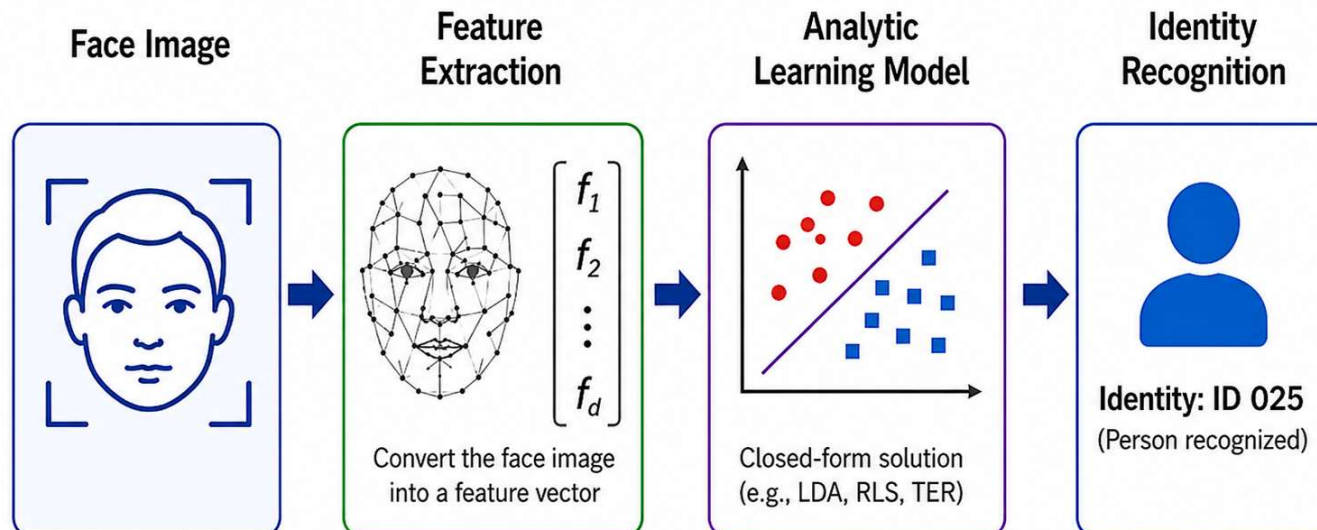
Learning becomes solving a mathematical problem rather than performing repeated optimization.

Representative Analytic Learning Algorithms

Course Topic	Representative Methods	Related Lecture
Linear Parametric Models	Least Squares Regression	L3
Learning Score Functions	Analytic Score Learning	L4
Analytic Regression	Ridge Regression / Regularized Least Squares	L5
Analytic Classification	Multi-class Analytic Classifiers	L6
Analytic Methods for Penalized Learning	Analytic Penalized Learning	L7

These analytic learning methods will be studied systematically in Lectures 3–7.

Face Recognition Using Analytic Learning



Efficient closed-form learning enables fast and accurate face recognition.

Efficient closed-form learning enables fast and accurate face recognition.



Figure 2: Samples of ten identities drawn from the CMU face data set.

Note: This dataset was created by Tom Mitchell. The dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license (<https://creativecommons.org/licenses/by/4.0/legalcode>).

The dataset is available at

<https://archive.ics.uci.edu/dataset/124/cmu+face+images>

- In this example, the task is to recognize images of faces captured under variations in pose, expression, and the presence of glasses.
- The Fig. shows several face image samples drawn from the database of CMU face images.
- This database contains 640 face images from 20 subjects with each subject contributing 32 different face images. Each face image of full resolution has 120×128 pixels, with 256 grey levels per pixel.
- Face recognition plays a crucial role in various applications, ranging from highly accurate custom and banking systems to more.
- Analytic learning models can be built to do face recognition, to be discussed in more details in subsequent lectures.

Advantages and Limitations of Analytic Learning

Advantages:

- Fast training
- Deterministic solutions
- Few hyperparameters
- Good interpretability
- Computational efficiency

Limitations:

- Applicable only under suitable model assumptions
- Less flexible for some highly nonlinear problems
- Some methods require matrix inversion
- Scalability remains an active research topic

Analytic learning complements rather than replaces iterative learning.

Recent Developments in Analytic Learning

- Closed-form deep architectures
- Continual analytic learning
- Distributed and edge AI
- Large-scale analytic optimization
- Explainable analytic models
- Hybrid analytic–iterative learning

Analytic learning continues to evolve as an active research area.

Summary and Transition

- ✓ Machine learning is not limited to iterative optimization.
- ✓ Closed-form learning provides an efficient alternative.
- ✓ Analytic learning is mathematically elegant and computationally attractive.
- ✓ The next major paradigm is **Ensemble Learning**.

Can multiple learners perform better than a single learner?

Can we improve the performance of a single learning model?

Many machine learning algorithms perform well on some datasets but poorly on others.

We combine several learning models to obtain:

- higher accuracy
- better robustness
- improved generalization

instead of relying on only one model

This idea leads to ensemble learning.

What Is Ensemble Learning?

Ensemble learning is a machine learning paradigm that

- trains multiple learners
- combines their predictions
- produces a final decision

The overall system is called an ensemble.

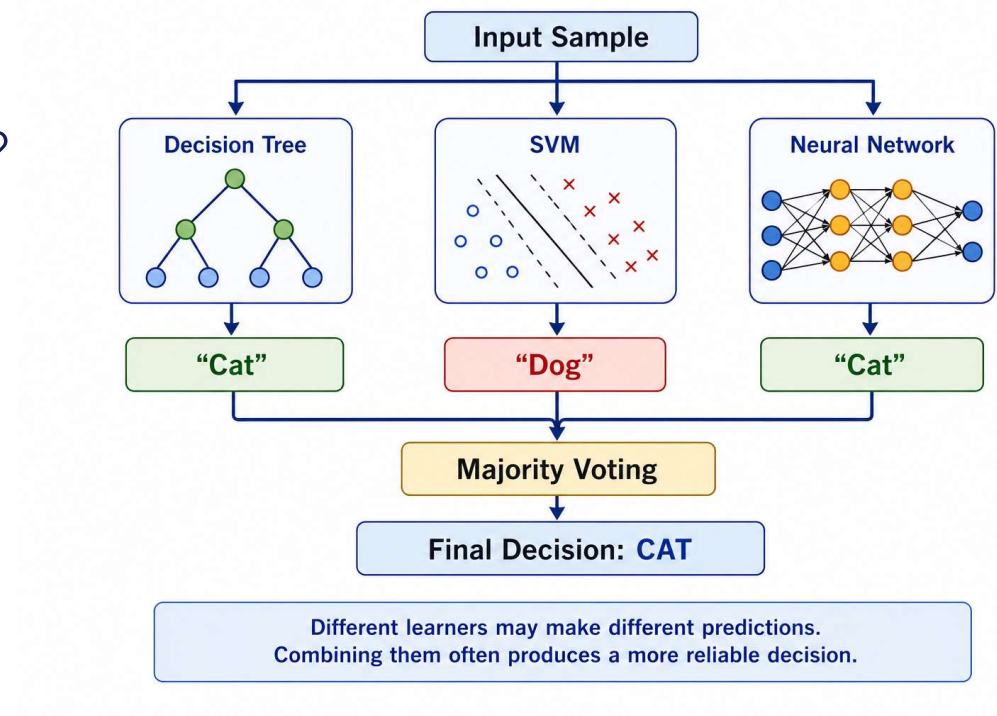
Key observation:

The whole can be better than the individual parts.

Why Does Ensemble Learning Work?

Three ingredients:

- Accurate learners
- Diverse learners
- Effective combination strategy



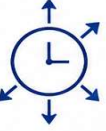
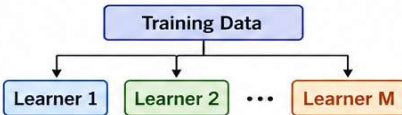
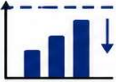
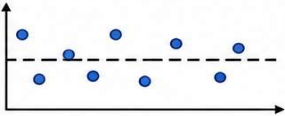
Successful ensemble learning requires both accuracy and diversity.

Basic Ensemble Strategies

Strategy	Basic Idea
Voting	Majority decision
Averaging	Average prediction
Weighted Combination	Learners contribute with different weights

Different ensemble algorithms mainly differ in how individual learners are combined.

Two Major Ensemble Strategies

Aspect	Bagging	Boosting
 Training Process	Parallel training 	Sequential training 
 Learners	Independent learners 	Dependent learners 
 Main Effect	Variance reduction 	Bias reduction 
 Characteristic	More robust Less sensitive to noise and overfitting	Higher predictive accuracy Can achieve lower bias and often better performance

Both methods have complementary strengths.

Representative Ensemble Learning Methods

Course Topic	Representative Methods	Related Lecture
 Introduction to Ensemble Learning	Bagging, Boosting	L9
 Classical Ensemble Algorithms	Random Forest, AdaBoost, Gradient Boosting	L10
 Advanced Boosting Algorithms	XGBoost, LightGBM	L11
 Reinforcement Learning with Ensemble Methods	Ensemble Reinforcement Learning	L12
 Industrial Applications	End-to-end Ensemble Models	L13

These ensemble learning methods will be studied systematically in Lectures 9–13.

Advantages and Limitations of Ensemble Learning

Advantages:

- ✓ Higher Prediction Accuracy
- ✓ Improved Robustness
- ✓ Reduced Overfitting
- ✓ Better Handling of Complex Problems
- ✓ Flexibility

Limitations:

- ✗ Higher Computational Cost
- ✗ Reduced Interpretability
- ✗ More Complex Model Management
- ✗ Diminishing Returns
- ✗ Dependent on Diversity

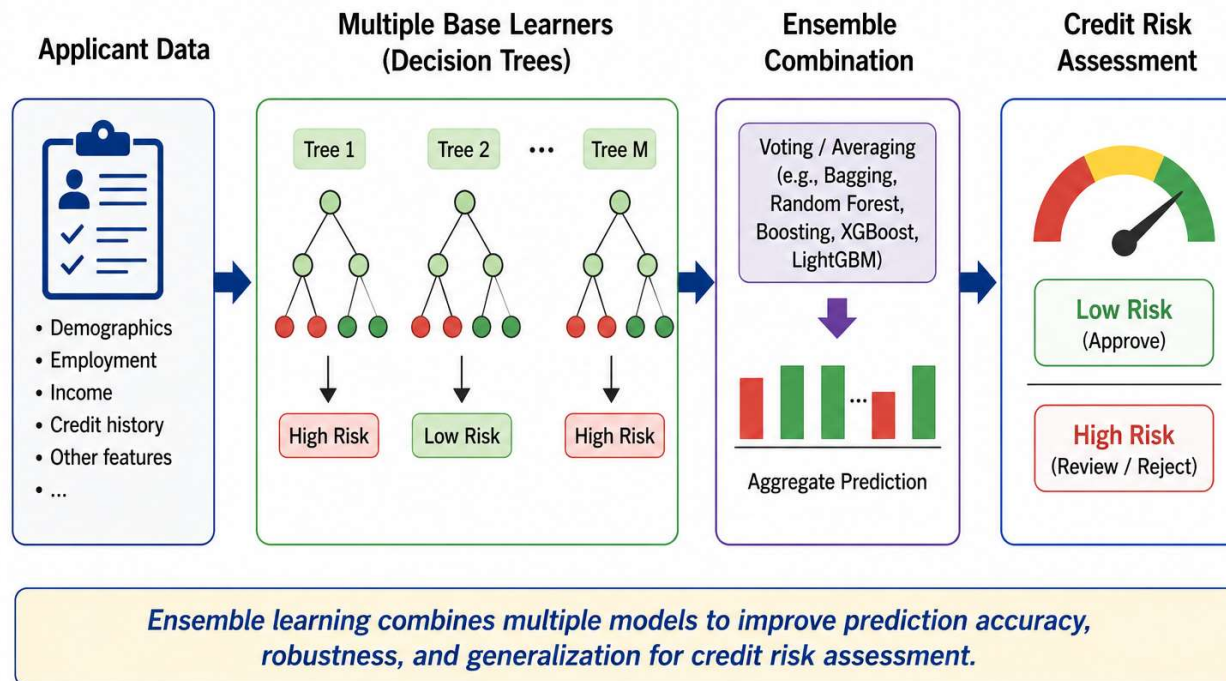
Ensemble learning improves prediction performance by combining multiple learners, at the expense of increased computational complexity and reduced interpretability.

Recent Developments in Ensemble Learning

- Deep ensemble learning
- Ensemble methods for foundation models
- Distributed ensemble learning
- Online and streaming ensembles
- Ensemble reinforcement learning
- Explainable ensemble models

Ensemble learning remains one of the most successful and active research areas in machine learning.

Credit Risk Assessment Using Ensemble Learning



Combining multiple learners improves prediction reliability and robustness.

Applications of Analytic and Ensemble Learning

Learning methods are now widely applied across many engineering and industrial domains.

Application Area	Typical Tasks	Typical Learning Methods
 Face Recognition	Recognize or verify a person from facial images	Analytic Learning (e.g., LSE, RLS, TER)
 Medical Diagnosis	Classify disease presence from medical data or images	Ensemble Learning (e.g., Random Forest, Boosting)
 Financial Risk Assessment	Evaluate credit risk, bankruptcy or default probability	Ensemble Learning (e.g., XGBoost, LightGBM)
 Industrial Inspection	Detect defects in products or components	Analytic & Ensemble Learning (e.g., TER + Boosting)
 Autonomous Driving	Perception, decision making and scene understanding	Ensemble Learning (e.g., Deep Ensembles, Boosting)
 Intelligent Manufacturing	Process monitoring, predictive maintenance, quality control	Analytic & Ensemble Learning (e.g., TER + Random Forest)



Key Message: Different learning methods are suitable for different application scenarios. The choice of learning method depends on the characteristics of the data and the application requirements.



Take-home message: *There is no single best learning algorithm for all problems.*

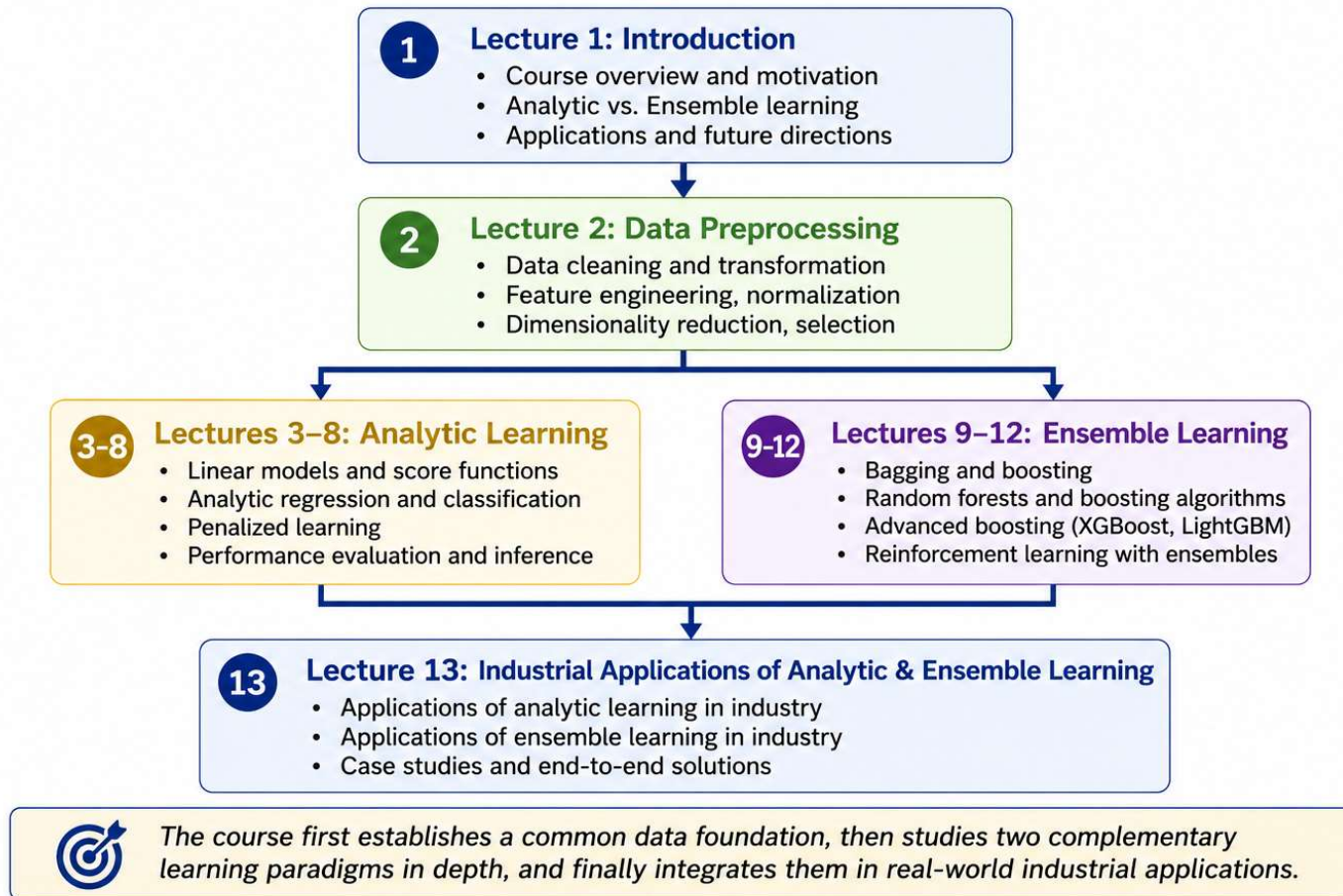
Future Directions in Machine Learning

- Emerging Research Trends
- Efficient Learning
 - Fast and scalable learning algorithms
- Explainable AI
 - Improving the transparency and interpretability of learning models
- Continual Learning
 - Learning new knowledge without forgetting previous knowledge
- Hybrid Learning
 - Combining analytic learning, deep learning and ensemble learning

Machine learning continues to evolve towards more efficient, intelligent and trustworthy systems.

EE6406 – Analytic and Ensemble Machine Learning

Course Roadmap



What You Will Learn in This Course

By the end of this course, you should be able to

- Understand the principles of analytic learning.
- Understand the principles of ensemble learning.
- Compare different machine learning paradigms.
- Select suitable learning methods for practical applications.
- Appreciate recent developments in modern machine learning.

The emphasis is on understanding principles rather than memorizing algorithms.

Key Takeaways

- Machine learning enables computers to learn from data.
- Analytic learning provides efficient closed-form solutions.
- Ensemble learning improves robustness by combining multiple learners.
- Different learning paradigms are suitable for different applications.
- This course focuses on both fundamental principles and practical methods.

Understanding the strengths and limitations of different learning paradigms is essential for developing effective intelligent systems.

Preview of Lecture 2

Data Preprocessing

Topics include:

- What is data?
- Data types
- Distance metrics
- Preprocessing methods

High-quality data is the foundation of successful machine learning.